DR-Net: A Multi-View Face Synthesis Network Driven by Dual Representation

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***Abstract*—With the rapid growth of virtual human techniques and the popularity of 3D face modeling, multi-view face synthesis has received considerable attention in the computer vision community. Though learning-based methods have achieved photo-realistic effect, the lack of geometry guidance remains a great hurdle to their results in high fidelity of identity. To relieve it, we propose a novel network named Dual-Representation-Net (DR-Net) driven by both learning and geometry information. In addition, we utilize face parsing to lead the learning of regional attention, in this way, the two synthesizing representations can collaborate and compete dynamically under different face angles, which improves the stability of face synthesis by assimilating the benefits of both kinds of information. What is more, we want to emphasize that DR-Net can serve as a general framework for multi-view face synthesis by jointly exploiting learning and geometry information. That is, our geometry module and learning module are actually substitutable, which can be replaced by any other SOTA methods. This makes our method both flexible and scalable. Extensive experiments verify that our approach is superior to other counterparts in both synthesis quality and identity preservation. Moreover, with the geometric information, we can decouple the synthesis of face and background to obtain more robust results on in-the-wild data.**

***Index Terms*—Multi-view face synthesis, 3D geometry modeling, deep learning, dual-representation net, attention network**

I. Introduction

Multi-view face synthesis has attracted increasing attention due to its wide applications in computer vision, face recognition, 3D object modeling and editing [1], [2]. Given a single-view face, the goal of this task is to generate images of the same person in multiple views, which requires the model to understand the complete object shape, not just the appearance under a single view. In this process, the identity characteristics and image attributes of the source image should be maintained in the generated images.

Traditionally, this task is addressed by geometry-based or deep learning-based methods. Based on the given single-view 2D face, the former reconstructs the 3D face model to obtain the geometry representation. By rotating the model to different target views, it is possible to generate renderings of multiple views through projection operations to the 2D plane [1], [3], [4]. Although these methods can maintain the identity characteristic of the face stably, the results only contain the inner part of the face, and the source image attributes (e.g. background, lighting and styles) and face attributes (e.g. hairstyle) must still be modeled separately. For example, Rotate-and-Render [1] uses a GAN-based translation network to map the rendered 2D face to the image domain with color background and hairstyle. However, it is inconsistent with the attribute conditions of the source image. Besides, due to the limited information of only one view, the face synthesis effect of the opposite side will be blurred when the input is a face on one side.

Recently, learning-based methods synthesize images by analyzing pixel-wise relationships to disentangle pose and identity features. Due to the superiority in generating realistic images, GAN is playing an increasingly important role through extracting common view-irrelevant features from the multi-view images [5]–[7]. Although these methods can maintain the face shape and the source image attributes to a certain extent, there are still some problems. (1) The facial features in the results output by the generator tend to be within the distribution range of the training data. (2) The result of facial parts are not comparable to geometry results. (3) These methods showing poor robustness on the wild data. To address the above problems, we design a novel network named Dual-Representation-Net (DR-Net) driven by both geometry and learning information. Specifically, given a single-view image, on the one hand, we obtain the geometry information through the 3D face model built by the single pose reconstruction method [8]; On the other hand, we use a two-pathway automatic encoder [5] framework to construct a face sketch under the target view, so as to get the learning representation. Driven by the two kinds of information, we design a region attention mechanism, which analyzes the quality of different face regions segmented by face parsing and assigns weights accordingly, so that they can mix in cooperation and competition to generate results, which draw on the advantages of both.

It is worthy of mentioning that DR-Net can serve as a general framework for multi-view face synthesis by jointly exploiting learning and geometry information, as our geometry module and learning module as well as the attention network are actually substitutable. That is, they can be replaced by any other existing methods or specifically designed modules. This indicates that DR-Net is both flexible to work with existing methods and scalable to embrace newly-developed models.

In summary, this paper has the following contributions: (1)We propose a novel framework that uses dual information representations of both geometry-based and learning-based to

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guide multi-view face synthesis, and combines the advantages of both to realize the simultaneous preservation of identity features and image attributes. (2)We design a region attention network guided by face parsing to assign weights to different face parts. The results based on geometry and learning are mixed accordingly so that they can compete and make up for each other. (3)Extensive experiments show the effectiveness and superiority of our method in synthesis quality and identity preservation. Furthermore, benefiting from the geometry information, we obtain more stable results on the wild data by decoupling the face from the background.

II. Related Work

View Synthesis. The geometry-based methods [8] obtained the rendered multi-view faces by rotating the 3D model to various target views and then projecting them onto the 2D plane. Rotate-and-Render [1] contains the internal face contents, hairstyles and image backgrounds, but the resulting hairstyle and image attributes are usually not consistent with the source image. Learning-based methods become increasingly popular with the help of encoder-decoder framework [9] to learn the view-invariant identity features. CR-GAN [5] extends this by introducing a generation sideway to learn the embedding space. ID-Unet [7] proposes an iterative framework that deforms the encoder features from different layers and gives them to the decoder to improve the synthesis quality. Xu et al. [10] synthesize multi-views by apply FFlowGAN multiple times. These learning methods can maintain image attributes, but they are not ideal in terms of maintaining facial identity features. In our work, we use both geometry and learning information, and implicitly learn the combination of the two through a specifically designed attention network.

Attention Networks. The attention mechanism has been widely used in the field of computer vision [11] including face-related tasks. Wang et al. [12] use an attention network to highlight the face area in the face detection task to solve the occlusion problem. NAN [13] uses a cascade attention mechanism for video face recognition, which compactly aggregates all frames in the video. RAN [11] uses the self-attention and relation-attention module to aggregate facial region features for facial expression recognition and combines the region-biased loss to strengthen the region weights. The attention mechanism can effectively capture the internal relationships of data, enabling the system to pay attention to key information and ignore irrelevant information. By taking each face region as a unit, we use a region attention network to learn the correlation (i.e., weights) between geometry-based and learning-based results, with which the final images are generated by weighted combination of the results of the geometry and learning branches.

III. Methodology

A. Overview

This paper aims to generate multi-view images when a single-view face is given while maintaining the image attributes and the identity characteristics of the face. Given the source image, we firstly use CRGAN [5] to generate a face sketch of the target view, in combination with the face landmark loss and weak feature matching loss. On the other hand, we reconstruct the 3d face model in the geometry module and render it to the 2D plane, and the pose is aligned with the output of the learning module. Furthermore, we design a region attention network, extract the features of each semantic face part and learn the corresponding weights by analyzing their quality performance. Finally, we obtain the images of comprehensive preserving the spatial features and image attributes according to the weights. The architecture of our DR-Net is shown in Fig. 1.

B. Learning Module

Given the source image I_S, we use a CRGAN [5] to generate a face sketch I_L under the target view. The two-pathway structure introduces a generation branch based on the single-pathway network, using random noise $\tilde{z}$ instead of the latent code $\overline{z}$ in the training set subspace to avoid the decoder G from overfitting the training samples.

Specifically, Given a face pair (x_i, x_j) with the same identity under the view v_i and v_j, in each iteration, the decoder G and the discriminator D are first trained according to the random noise $\tilde{z}$, so that the completed latent representation is learned. The decoder generates an image $\tilde{x}$ under view v_i according to $\tilde{z}$ and compares it with x_i. Then, the decoder G is fixed, and the encoder E and discriminator D are trained by reconstructing x_j from x_i. The encoder outputs an identity-preserved code $\overline{z}$ according to x_i, and the decoder generates the image $\overline{x}$ from the view v_j and compares it with x_j.

Although the two-pathway design has considered the overfitting of the training data, this learning-based method decouples the view and the identity information by simply analyzing the relationship between pixels, and lacks certain guidance of spatial information. When the input is external data with different backgrounds, it is difficult for the encoder to distinguish the face part from the unseen background, resulting in blurred images. In view of this, we introduce the 3D landmark information of the face. By measuring the facial landmark difference between the generated image and the ground-truth image, the network can pay attention to the edges and boundaries, so as to provide help to extract the face part from the "unseen" data and improve the structural consistency of the generated image. This landmark consistency loss $\mathcal{L}_{\mathrm{ldk}}$ is formulated as follows:

$$\mathcal{L}_{\mathrm{ldk}}(z, v, x) = \|L(G(z, v)) - L(x)\|_2 \tag{1}$$

where L is the pre-trained landmark detector, z is the latent code, and x is the real image under the target view v.

To maintain the image attributes of the source face, we use weak feature matching loss $\mathcal{L}_{\mathrm{wfm}}$, which evaluates the image by comparing the feature matching of multi-layer features in the discriminator [14]. Since the image attributes belong to the high-level semantic information in the deep features, we

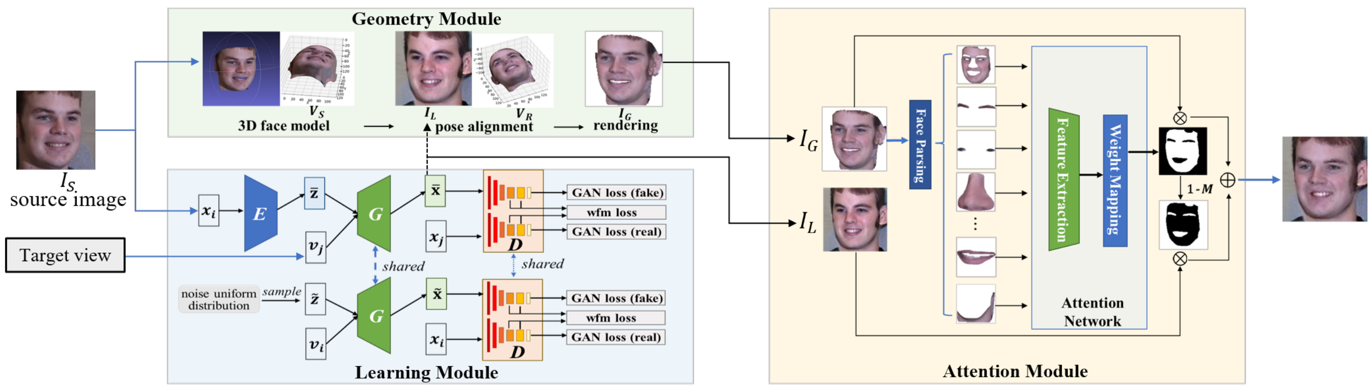

Fig. 1. The framework of our Dual-Representation-Net (DR-Net) .

constrain the attributes by matching the features of the later layers:

$$\mathcal{L}_{\text{wfm}}\left(x_G, x\right)=\sum_{i=q}^{Q} \frac{1}{e_i}\left\|D^i\left(x_G\right)-D^i(x)\right\|_1 \tag{2}$$

where x_G is the output of the generator and x is the ground-truth image. Q denotes the total number of layers, q is the starting layer label included in the calculation, and e_i refers to the number of elements on the i-th layer. During training, we also use the adversarial loss $\mathcal{L}_{\text{adv}}$ in [5].

Therefore, the overall loss can be written as:

$$\mathcal{L}_{\text{learning}}=\lambda_{\text{ldk}} \mathcal{L}_{\text{ldk}}+\lambda_{\text{wfm}} \mathcal{L}_{\text{wfm}}+\lambda_{\text{adv}} \mathcal{L}_{\text{adv}} \tag{3}$$

After the completion of training, given the source face x_i, the result of the generator under the target view v_j is the output I_L of the learning module.

C. Geometry Module

Given the single-view source face I_S, we obtain the 3D face model based on 3DDFA [8] and the set $V_S \in R^{3\times n}$ of 3D vertex coordinates corresponding to the positions in I_S, which strictly follows the facial features of the current face itself. In order to facilitate subsequent mixing, we need to rotate the face to align the pose with the output I_L of the learning module.

Through pose estimation [15], we can get the Euler angles $\boldsymbol{\theta}_S$ and $\boldsymbol{\theta}_L$ corresponding to the source image I_S and the output I_L of the learning module, where Euler angles are the rotation angles of the face around the three x, y, z coordinate axes and are denoted as $\boldsymbol{\theta}=(\theta_x, \theta_y, \theta_z)$. Then, we can use the difference between the two to calculate the rotation angle $\Delta\boldsymbol{\theta}$ required to rotate the 3D vertex of I_S to the posture of I_L in the 3D coordinate system, and this can be formalized as:

$$\Delta\boldsymbol{\theta}=\boldsymbol{\theta}_S-\boldsymbol{\theta}_L \tag{4}$$

According to the rotation angle $\Delta\boldsymbol{\theta}$, we construct the 3D rotation matrix, multiplied by the original coordinates V_S, and obtain the rotated coordinates. An offset Δb is added to align the coordinates with the learning module output I_L. The rotated coordinates V_R are calculated as follows:

$$V_R=\begin{bmatrix} c_y c_z & -c_y s_z & s_y \\ s_x s_y c_z+c_x s_z & -s_x s_y s_z+c_x c_z & -s_x c_y \\ -c_x s_y c_z+s_x s_z & c_x s_y s_z+s_x c_z & c_x c_y \end{bmatrix} V_S+\Delta b \tag{5}$$

where $c_i=cos\left(\Delta\theta_i\right)$, $s_i=sin\left(\Delta\theta_i\right)$ and $i\in\{x,y,z\}$.

We render the rotated coordinates V_R to the x-y plane, thus the face I_G of the target view can be obtained under the guidance of the geometric information of the 3D model.

D. Attention Module

After obtaining the output I_L of the learning module and the output I_G of the geometry module, in this section, we focus on using the strengths of each to generate the target view image. The results are expected to highly maintain the identity characteristics while following the source image attributes. Due to their advantages and disadvantages in different facial parts, we design a region attention network to extract the features of different parts. By assigning weights to them and mixing I_L and I_G accordingly, we can get the final image.

Specifically, according to semantic attributes, we first use the face parsing [16] method to segment the face I_G into k parts (e.g. skin, eyebrows, eyes, nose, mouth, and neck). The corresponding mask matrix is denoted as $M_1, M_2, \cdots, M_k$, where M_i is the same size as the image pixel matrix, and the value of the position belonging to the current part is 1, the rest is 0, and the cropped images are denoted as $P_1, P_2, \cdots, P_k$.

To learn the weights of each part, these parts are fed into a convolutional neural network (CNN) to extract their features, so as to judge the quality of each facial part in I_G. Let $f_\varphi(\cdot)$ represent the CNN with φ as the parameter, then the feature F_i of each part of CNN output can be denoted as:

$$F_i=f_\varphi\left(P_i\right)\ \ (i=1,2,\ldots,k) \tag{6}$$

Followed by a fully connected (FC) layer and a sigmoid function, the attention network can get the quality mapping value of each part. If it is greater than the threshold τ, the weight α_i of this part is 1 and the cropped part will be adopted in the final image. Let W_{fc} be the FC parameter, then:

$$\alpha_i=H\left(Sigmoid\left(W_{fc}^T * F_i\right)-\tau\right) \tag{7}$$

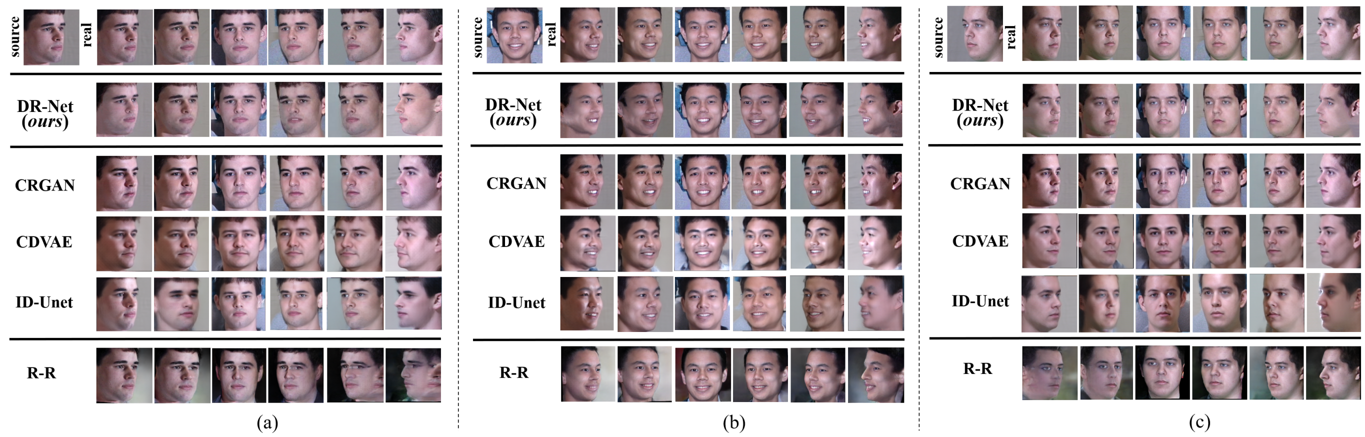

Fig. 2. Four sets of visual comparison examples of our method (DR-Net) with the learning-based methods (CRGAN, CDVAE and ID-Unet) and the geometry-based method (Rotate-and-Render (R-R)). In each group, the left is the input image, and the right sequence is the synthesis results of views ranging from -60° to +60°.

where $H(\cdot)$ is the step function. Based on the weights, we can get the mask matrix M when I_L and I_G are mixed:

$$M = \sum_{i=1}^{k} \alpha_i M_i \tag{8}$$

Thus, the final image I_F is obtained by:

$$I_F = (1 - M) \odot I_L + M \odot I_G \tag{9}$$

where $\odot$ is Hadamard product. In this way, the final image I_F can not only have the background of I_L, but also select the better composition result according to the quality performance of I_L and I_G in each facial semantic part.

In the training process, CNN with shared weights in different facial parts and the fully connected network need to be trained to output appropriate masks to get mixed I_F. We use the identity preservation loss $\mathcal{L}_{\text{id}}$ to make I_F follow the identity feature of the ground-truth image as close as possible under the target view. It is formulated as:

$$\mathcal{L}_{\text{id}} = 1 - cos\left(h\left(I_F\right), h\left(I_{gt}\right)\right) \tag{10}$$

where $h(\cdot)$ is the pre-trained face recognition model to output the face feature vector, I_{gt} is the ground-truth image of the current identity under the target view, and $cos(\cdot, \cdot)$ represents the cosine similarity between the two vectors.

We also use the pixel-level loss to make I_F close to the texture details of I_{gt}, that is:

$$\mathcal{L}_{\text{pix}} = \frac{1}{2}\left\|I_F - I_{gt}\right\|_2 \tag{11}$$

Finally, the attention network is trained to minimize the weighted sum of the above losses:

$$\mathcal{L}_{\text{attention}} = \lambda_{\text{id}}\mathcal{L}_{\text{id}} + \lambda_{\text{pix}}\mathcal{L}_{\text{pix}} \tag{12}$$

IV. Experiments

A. Experimental Settings

a) Datasets and Compared Methods: **Multi-PIE** [17] is a dataset of 130,000 images containing 250 identities collected under 13 viewing angles spanning 180 degrees, 20 illuminations, and two expressions. **CelebA** [18] contains 202,599 wild faces covering various poses of 10,177 celebrities crawled from the web. We conduct experiments on the Multi-PIE dataset and verify the generalization of our method on CelebA. We compare our method with four typical existing counterparts, including three learning-based methods: CRGAN [5], CDVAE [6], and ID-Unet [7], and one state-of-the-art geometry-based method Rotate-and-Render (R-R) [1].

b) Metrics: we use the LPIPS, FID and SSIM metrics to evaluate the performance quantitatively. To verify the identity feature preservation of the generated images, we use the face recognition model ArcFace [19] to generate face embeddings of the images, and then the cosine distance between source and real images is calculated to measure the identity similarity (ID-sim). The quantitative results are shown in Table III-D.

c) Implementation Details: our DR-Net consists of three modules, which need to be trained separately. For the geometry module, we follow the training method of 3DDFA [8] to get a 3D face reconstruction network. For the learning module, our network implementation is optimized based on the two-pathway network in CRGAN [5]. During training, the value of v is taken from 9 viewing angles and the latent code $z \in [-1, 1]^{119}$. The values of λ_{ldk}, λ_{wfm}, and λ_{adv} are set to make the values of each loss item balanced. The learning rate of adam optimizer is 0.0001 and the momentum is [0, 0.9]. λ_{pix} and λ_{id} are set to 0.1 and 20 respectively. For the attention module, we use the ResNet18 as the backbone CNN. The initial learning rate is 0.01, divided by 10 after 12, 20, and 32 epochs. The maximum number of training epochs is 40. All the models are trained using Pytorch on 4 Tesla V100 GPUs with 16 GB memory.

B. Qualitative and Quantitative Comparison Results.

Results on the Multi-PIE Datasets. We first evaluate DR-Net on the Multi-PIE dataset. We select six views within -60° to +60° for face synthesis.The qualitative results of our DR-Net and comparisons with existing methods are shown

TABLE I
PERFORMANCE COMPARISONS ON MULTIPIE AND CELEBA AMONG OUR DR-NET, 3 LEARNING-BASED METHODS (CRGAN, CDVAE AND ID-UNET) AND THE SOTA GEOMETRY-BASED METHOD (ROTATE-AND-RENDER (R-R)). THE ARROW INDICATES THE DESIRED VALUE PREFERENCE.

Method	MultiPIE					CelebA				
	$L_{\text{pix}} \downarrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	ID-sim $\uparrow$	$L_{\text{pix}} \downarrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	ID-sim $\uparrow$
CRGAN	21.47	0.632	25.65	0.152	0.613	41.49	0.332	64.69	0.244	0.105
CDVAE	21.32	0.619	25.47	0.147	0.632	59.02	0.305	52.12	0.267	0.141
ID-Unet	20.63	0.667	**23.98**	0.121	0.802	39.17	0.436	59.48	0.213	0.327
R-R	48.55	0.493	41.89	0.216	0.795	48.51	0.337	55.19	0.240	0.674
DR-Net	**20.09**	**0.725**	25.34	**0.107**	**0.864**	**24.92**	**0.504**	**44.46**	**0.187**	**0.691**

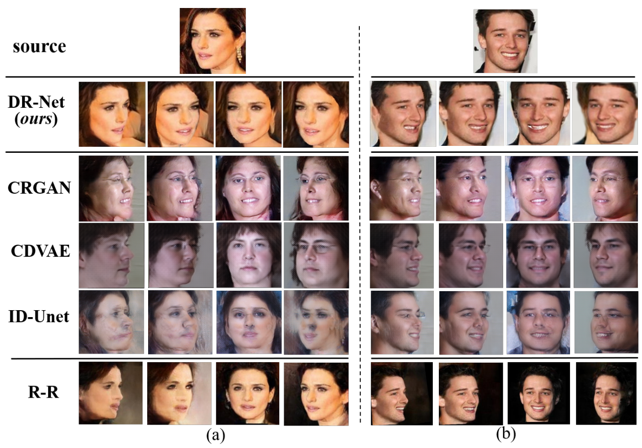

Fig. 3. Two sets of qualitative comparisons on CelebA to verify the generalization of different methods.

in Fig. 2. We can see that the identity features of learning-based methods are over-fitting to the training data domain in all views, and the image style of geometry-based method(R-R) is not consistent with the source image. Specifically, CRGAN [5], CDVAE [6], and ID-Unet [7] are not realistic due to their dependence on the training samples. For example, in Fig. 2(c), the eye' color of the identity is blue, but the majority of the training samples are black, so the results obtained are all black. On the other hand, Rotate-and-Render (R-R) [1] uses the same information from the reconstruction model as we do. The 3D information in R-R is only used to generate the interior of the face, and the image background and hairstyle need to be modeled separately, which cannot follow the style of the source image well. Our results preserve the illumination and environment satisfyingly. Quantitative results in Table III-D also demonstrate that our method is optimal in terms of pixels (L_{pix}), structure (SSIM), and perception distance (LPIPS). We also achieve competitive results on FID. The geometry-based method Rotate-and-Render has the worst performance in these three indicators because it does not comply with the attributes of the source image.

a) Results on the Wild Dataset CelebA.: With the trained model on Multi-PIE, we verify the generalization of our method by randomly selecting 1000 images from CelebA. Qualitative results in Fig. 3 demonstrate that our method keeps the identity characteristics consistent with the source image. For the learning-based methods, the image attributes and background of generated images in all perspectives show a "Multi-PIE style", their maintenance of identity features is not ideal. The geometry-based method R-R maintains the identity characteristics well, but the background of the image and the hairstyle of the person are different from the source image. these existing methods rely only on the analysis of the relationship between pixels to extract identity representations, but when facing the wild data, it is difficult to find the original pixel relationship to obtain good latent representations. Our method introduces the landmark loss and weak feature matching loss in the learning module to provide spatial guidance for feature extraction and uses the information of the geometry module to make corrections. For quantitative evaluation, 1000 images are selected to generate corresponding multi-view images, and then the generation results of the same view as the source image are compared with the source for metric calculation. Because the learning-based methods (CRGAN, CDVAE, and ID-Unet) overfit the distribution domain of Multi-PIE, and the geometry-based method (R-R) does not comply with the attributes of the source image, their performances of L_{pix}, SSIM, LPIPS, and FID are inferior to ours.

TABLE II
QUANTITATIVE RESULTS OF THE ABLATION STUDY.

Method	$L_{\text{pix}} \downarrow$	SSIM $\uparrow$	FID $\downarrow$	LPIPS $\downarrow$	ID-sim $\uparrow$
A: unoptimized setting	41.49	0.332	64.49	0.244	0.105
B: A + $\mathcal{L}_{\text{wfm}}$	30.33	0.412	45.62	0.205	0.239
C: B + $\mathcal{L}_{\text{ldk}}$	28.53	0.447	**43.33**	0.198	0.287
D: C + geometry ($\mathcal{L}_{\text{id}}$)	25.80	0.475	49.71	**0.182**	0.648
E: D + $\mathcal{L}_{\text{pix}}$	**24.92**	**0.504**	44.46	0.187	**0.691**

C. Ablation Study

To validate the necessity of each component in our method, we conduct an ablation study on CelebA and demonstrate the quantitative results in Table II. Due to page limitations, qualitative results are shown in supplementary materials. Specifically, (1) In **setting A**, we use the unoptimized learning module, which does not include weak feature matching loss $\mathcal{L}_{\text{wfm}}$ and landmark loss $\mathcal{L}_{\text{ldk}}$, and the result is weak in maintaining image attributes and identity features on the wild data. (2)

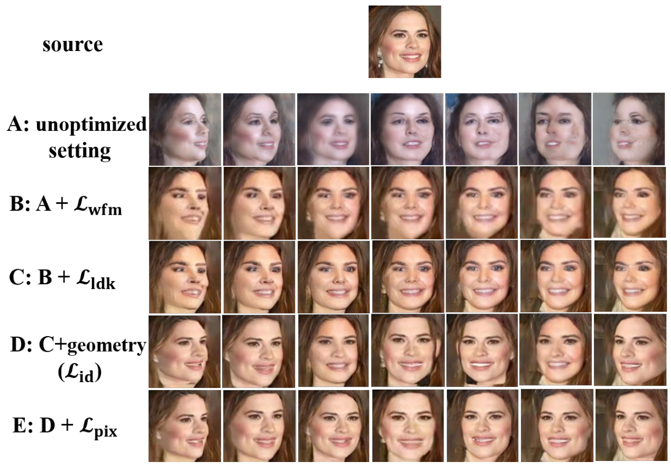

Fig. 4. Qualitative results of the ablation study.

Setting B adds loss $\mathcal{L}_{\mathrm{wfm}}$ to constrain the attributes. From the visual and numerical results, it can be seen that the preservation of image attributes has been significantly improved, and the errors on pixels and feature levels (LPIPS) have been significantly reduced by the ratio of 26.89% and 15.98%. (3) **Setting C** further adds landmark loss $\mathcal{L}_{\mathrm{ldk}}$ to verify the role of this spatial guidance. The results show that the contours of human faces are more distinct, and the identity similarity (ID-sim) is also improved. (4) In **setting D**, we start to introduce geometry information, and use id loss $\mathcal{L}_{\mathrm{id}}$ to train the attention network. The geometry information significantly increases ID-sim from 0.287 to 0.648, but the FID result drops slightly by 6.38. (5) Finally, we test the effect of pixel loss $\mathcal{L}_{\mathrm{pix}}$ in **setting E**, and the results show that it alleviates the influence of geometric information introduction on FID and improves identity similarity.

In D of Figure 4, using $\mathcal{L}_{\mathrm{id}}$ alone may lead to problems such as inconsistent skin color (D_3, D_6) and unnatural fusion of neck and cheek edges (D_1, D_4, D_5, D_7), which are solved by the addition of $\mathcal{L}_{\mathrm{pix}}$. The above results show that all the components we use are critical to the performance of the method.

V. Conclusion

In this paper, we propose a novel network named DR-Net that uses dual information representations of both geometry-based and learning-based to guide multi-view face synthesis. Through the region attention mechanism, we implicitly learn how they can collaborate and compete dynamically under different angles, which obviously improves the stability of face synthesis while absorbing the benefits of both kinds of information. Extensive experiments on both training and in-the-wild data demonstrate that DR-Net has superior synthesis quality, stability, as well as the ability of identity preservation over other counterparts.

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